

Supplementary Information

Composable Assurance for Sovereign Digital Currency: Privacy, financial integrity, adaptive policy and national-scale qualification across independent sovereign ledgers

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S1. Evidence taxonomy and reproducibility boundary

The evidence hierarchy separates formal or exhaustive state exploration, synthetic benchmark evidence, single-host laboratory prototype measurements, trace-driven digital-twin results, physical multi-region qualification and independently observed pilot evidence. A claim may not be promoted to a higher evidence class without the corresponding test.

The current package supports bounded protocol safety, synthetic privacy and AML/CFT findings, sanctions-ranking calibration, conditional economic exposure analysis and simulation-based capacity falsification. It does not support a national deployment claim.

S2. Monetary state and cross-border certificate protocol

Wholesale issuance transforms reserve liabilities into CBDC inventory. Retail conversion exchanges a commercial-bank deposit for existing inventory. Supply conservation is represented by $C = \text{sum}(I_b) + \text{sum}(W_u) + E + O + A$, where C is issued CBDC, I_b bank inventory, W_u user wallets, E settlement escrow, O risk-bounded offline value and A explicit system accounts.

The cross-border protocol binds payment intent, unequal-currency FX quote, bilateral compliance authorizations and final domestic prepare receipts. For seven signers and quorum five, any two quorums intersect in at least three nodes; conflicting terminal certificates are excluded under the stated bound of at most two equivocating keys and durable one-decision locks for honest signers.

No domestic ledger releases escrow because of silence or local timeout. Finalization requires a valid COMMIT certificate; release requires a valid ABORT certificate.

S3. Privacy attacker model and learned red-team design

The attacker classes are the ledger operator, network observer, merchant coalition and compromised PIP. Features include timing, endpoint classes, amount bands, merchant repetition, wallet implementation fingerprints and behavioural summaries. Evaluation uses unseen-user splits, an independently generated behavioural regime and ranking sets up to 3,000 candidates.

The formal linkability objective is $\text{Adv_link}(A) = |\Pr[A(L_0, L_1) = b] - 1/2|$. The empirical experiment intentionally relaxes ideal traffic assumptions and measures residual metadata leakage. The shielded and batched design materially reduces independent-generator network-observer AUC but does not eliminate unseen-user linkage.

S4. Registered person-level differential privacy

Two datasets are adjacent when one person and all that person's registered contributions are added or removed. A person contributes to at most three histogram cells per monthly release, giving L1 sensitivity three. The mechanism is $M(D) = \max(0, f(D) + \text{Laplace}(3/\epsilon_t))$. Twelve releases at $\epsilon_t = 1/3$ yield annual epsilon four under sequential composition.

The guarantee applies only to the registered public-statistics service and its accountant. It does not cover raw curator data, AML graphs, identity mappings, case files or arbitrary dashboard queries. Sparse-cell bias and clipping are reported as utility costs.

S5. AML/CFT predictive evaluation and prospective effectiveness

Sixteen independently generated graphs are drawn from four families: profile mix, calendar/payroll, merchant network and remittance corridor. The PIP-local baseline uses only transactions visible to the institution. Network features are rebuilt independently inside training and test subgraphs, and the graph is the unit of statistical inference.

The mean network PR-AUC improvement is reported with graph-level bootstrap confidence intervals and non-parametric paired tests. The evaluation demonstrates heterogeneous predictive improvement, not FATF effectiveness. Prospective validation must measure alert disposition, evidence sufficiency, investigator time, reporting quality, FIU feedback, disruption outcomes, false harm and subgroup effects.

S6. Sanctions-screening calibration and workload

The benchmark indexes 19,178 official records and 20,010 aliases. Positive entities are disjoint across training, calibration and test. Perturbations cover exact forms, official aliases, OCR corruption, token order, transliteration and sparse attributes. Negative identities and hard near matches are generated, which limits external validity.

Thresholds are selected on a calibration set for high-recall, balanced and low-workload policies. Results are converted from balanced-benchmark precision to prevalence-adjusted positive predictive value and analyst workload. At 0.01% assumed prevalence, even a low false-positive rate can generate thousands of alerts per million screenings.

The score is candidate ranking, not a legal determination. Ownership, programme rules, identifiers, transaction facts, licences and analyst review remain necessary.

S7. Economic partial identification and adaptive policy

Let $\theta=(H,F,r)$ contain a holding threshold, conversion-flow limit and remuneration rule. The feasible policy set contains parameters satisfying bank-liquidity, funding, ordinary-payment utility, inclusion, distributional and legal constraints over an uncertainty set U . Policy selection minimizes expected loss and tail risk rather than assuming a single transferable optimum.

An exposure ceiling $H \leq \tau D/(N a)$ is a conditional upper bound, where τ is tolerated gross deposit exposure, D eligible deposits, N eligible users and a adoption share. It is not an optimality equation. Jurisdictional estimation requires household demand, bank replacement funding, liquidity-facility use, deposit repricing, subgroup outcomes and crisis scenarios.

S8. National-scale qualification and queueing model

The analytical capacity formula $n_j \geq \lceil \lambda h_{s_j} / \rho_{\max} \rceil$ is a necessary lower bound for stage j , not a hardware guarantee. The trace-driven tandem-queue model includes API/authentication, policy and sanctions, HSM verification, monetary transition and durable audit stages under bursts, HSM slowdown and regional loss.

A national performance claim requires physical tests across at least three failure domains, certified HSMs, realistic open-loop and closed-loop PIP workloads, sustained load at least twice observed peak, simultaneous component failures, RTO/RPO evidence, no monetary-invariant breach or split finality, independent observation and controlled pilot exit criteria.

S9. Reproduction instructions and artifact inventory

Install dependencies from requirements.txt, then execute RUN_ALL.sh followed by VERIFY.sh. The artifact includes scripts, public and generated data, results, figures, documentation, checksums, licences and an immutable Git bundle. The included commit is 6fc7fa1ce098873b81fdb9dedb4e097c72727b93.

Because a public repository and DOI have not yet been assigned, the artifact should be deposited in a persistent repository before final submission. The deposited release must match the submitted checksum.

S10. Supplementary limitations and deployment gates

Privacy and AML/CFT experiments remain synthetic. The sanctions negative population does not reproduce all languages, ownership structures or customer-data quality problems. The economic model does not estimate all behavioural parameters from confidential microdata. The queueing model cannot represent every production interaction. The cross-border model excludes indefinite denial of service, dynamic membership and complete post-compromise recovery.

Required external gates include governed real-data studies, multilingual analyst adjudication, structural economic estimation, formal protocol refinement, independent cryptographic review, certified HSM testing, physical multi-region qualification, legal opinions and controlled central-bank pilots.

S11. Selected supplementary result tables

Privacy learned attacks

seed	profile	attacker	split	pairwise_auc
0	rotation_only	network	unseen_users	0.9924640624999999
0	rotation_only	network	independent_generator	0.9487937499999999
0	rotation_only	compromised_pip	unseen_users	0.9922140624999999
0	rotation_only	compromised_pip	independent_generator	0.9480859374999999
0	relay_standardized	network	unseen_users	0.917434375
0	relay_standardized	network	independent_generator	0.7362328125
0	relay_standardized	compromised_pip	unseen_users	0.9239687499999999
0	relay_standardized	compromised_pip	independent_generator	0.7615609375000001
0	shielded_batched	network	unseen_users	0.8134453125000001
0	shielded_batched	network	independent_generator	0.6154828124999999
0	shielded_batched	compromised_pip	unseen_users	0.8343578125
0	shielded_batched	compromised_pip	independent_generator	0.6614781249999999
1	rotation_only	network	unseen_users	0.990709375
1	rotation_only	network	independent_generator	0.9507109375
1	rotation_only	compromised_pip	unseen_users	0.9906687499999999
1	rotation_only	compromised_pip	independent_generator	0.9514640625
1	relay_standardized	network	unseen_users	0.934678125
1	relay_standardized	network	independent_generator	0.74531875
1	relay_standardized	compromised_pip	unseen_users	0.9377765625

Source: reproducibility artifact result file privacy_v7_learned_attack.csv

AML graph-level paired effects

dataset_id	family	network	pip_local	delta
0	profile_mix	0.5228687270811372	0.44153008437216534	0.08133864270897184
1	profile_mix	0.4770393083787755	0.5410780656904057	-0.06403875731163022
2	profile_mix	0.5674139597224512	0.5779586127607802	-0.010544653038328966
3	profile_mix	0.5778067306509204	0.4371614056976888	0.14064532495323157
4	calendar_payroll	0.413525854923585	0.3518913475334998	0.06163450739008519
5	calendar_payroll	0.4016523954854262	0.33952565017653275	0.06212674530889345
6	calendar_payroll	0.441844051945228	0.33841903678816987	0.10342501515705815
7	calendar_payroll	0.5880624464751003	0.5031774377934157	0.08488500868168458
8	merchant_network	0.3780835145038755	0.301696626561151	0.07638688794272452
9	merchant_network	0.2932519235907404	0.40058809479637014	-0.10733617120562972
10	merchant_network	0.4992062235116068	0.4291530986405133	0.0700531248710935
11	merchant_network	0.44832392909236035	0.459350600304954	-0.011026671212593653
12	remittance_corridor	0.4192161252897497	0.3974834051436873	0.021732720146062423
13	remittance_corridor	0.5789551085871513	0.45733833804145635	0.12161677054569492
14	remittance_corridor	0.6062684025562595	0.42276998039177416	0.1834984221644853
15	remittance_corridor	0.33973771071573755	0.22654425956297264	0.11319345115276491

Source: reproducibility artifact result file aml_v7_paired_effects.csv

Sanctions calibrated policy points

threshold	precision_ben chmark	recall	fpr	tp	fp	tn	fn
0.59956632314 39466	0.80779220779 22078	0.95692307692 30769	0.05285714285 714286	311	74	1326	14
0.73134167083 54177	0.98269896193 77162	0.87384615384 61538	0.00357142857 14285713	284	5	1395	41
0.9	0.99595141700 40485	0.75692307692 30769	0.00071428571 42857143	246	1	1399	79

Source: reproducibility artifact result file sanctions_v7_policy_points.csv

Economic identified ceilings

maximum_gross_exposure_share	eligible_users_million	adoption_share	implied_holding_limit_ceiling_eur
0.03	250	0.25	4733.2571232
0.03	250	0.5	2366.6285616
0.03	250	0.75	1577.7523744
0.03	250	1.0	1183.3142808
0.03	300	0.25	3944.380936
0.03	300	0.5	1972.190468
0.03	300	0.75	1314.7936453333334
0.03	300	1.0	986.095234
0.03	358	0.25	3305.3471530726256
0.03	358	0.5	1652.6735765363128
0.03	358	0.75	1101.7823843575418
0.03	358	1.0	826.3367882681564
0.05	250	0.25	7888.761872
0.05	250	0.5	3944.380936
0.05	250	0.75	2629.587290666667
0.05	250	1.0	1972.190468
0.05	300	0.25	6573.968226666667
0.05	300	0.5	3286.9841133333334
0.05	300	0.75	2191.322742222222

Source: reproducibility artifact result file economic_v7_identified_ceiling.csv

National qualification scenarios

scenario	arrival_target_tps	observed_arrival_tps	domestic_p50_ms	domestic_p95_ms	domestic_p99_ms	cross_border_p99_ms	region_loss
TIPS-reference steady	2000	1995.15	36.942586944488916	59.775921546003886	73.2449755754315	298.0155750831808	False
FuSSE-reference steady	10000	9981.6	36.95609332849298	59.956423499527496	73.51001805893524	298.5822703084517	False
FuSSE-reference burst	10000	9019.3	36.92282381637476	59.91424137612651	73.19403549373438	300.47787364230396	False
10k with one region unavailable	10000	9721.45	63.75583770188936	104.75327246145584	128.60101807722623	480.890001477112	True
10k with HSM service doubled	10000	9922.7	37.23077647956491	60.20279604688046	73.72312115854521	300.6920801090211	False
20k qualification stress	20000	18616.333333333332	37.48771822088869	60.481249973541054	73.97055088124569	301.7218512968453	False
40k overload control with region loss	40000	36607.875	4248.858340615166	8071.231940455444	8432.220293145741	479.7509915664931	True

Source: reproducibility artifact result file performance_v7_scenarios.csv